HUMANSCALE

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CAD Workshop-2: AI-aided CAD Design

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Objective

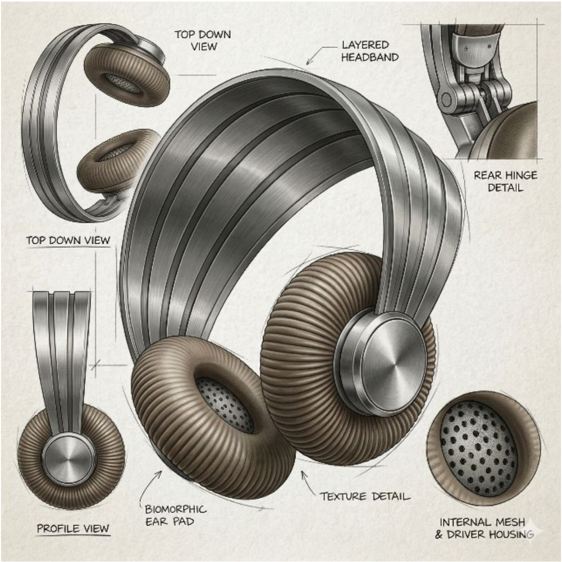
1.Build draft 3D models from 2D images using AI, and evaluate their topology for further modification.

2.Use Blender to fit the 3D models onto physical 3D scan data.

3.AI-assisted edit and refine the 3D models.

4.Render the models and explore other post-processing operations.

Sketch generation and normalization



Example prompt:

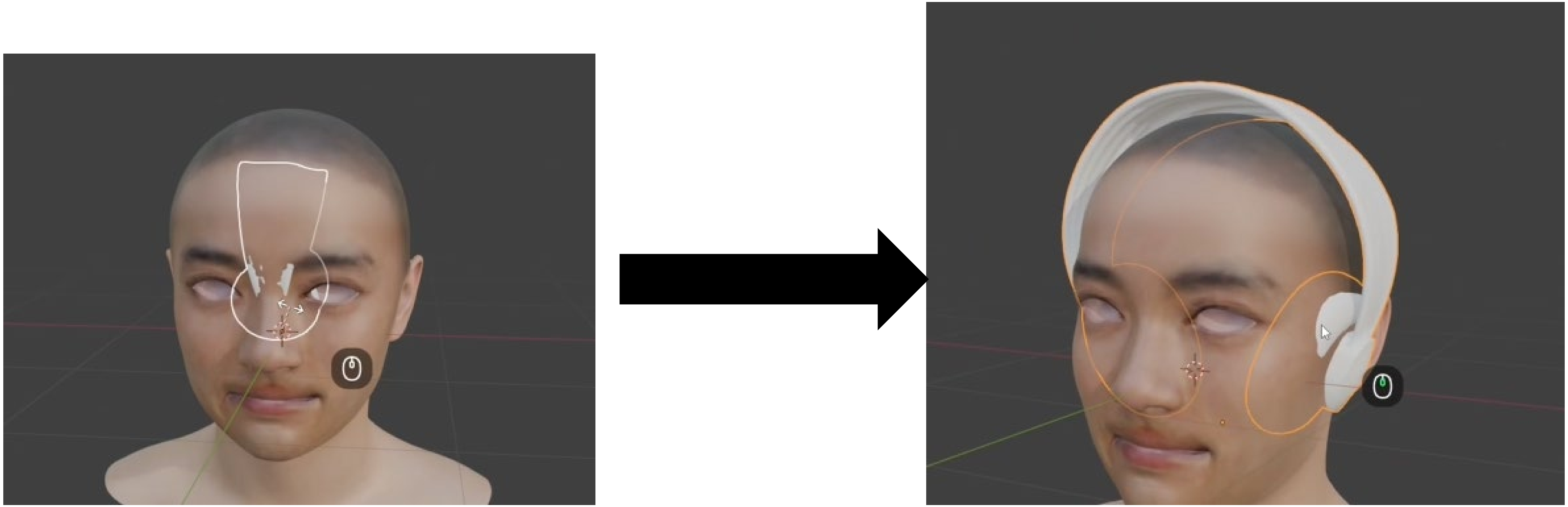
Act strictly as a commercial product renderer. Your sole task is to generate a studio-quality product render based on the provided reference image without altering its underlying geometry. You must strictly preserve the exact shapes, structural boundaries, and all design details of the reference object. Do not hallucinate, add, modify, or redesign any geometric features. Render a photorealistic [Subject, e.g. retro-futuristic over-ear headphones featuring a ribbed headband and textured white ear cushions] from a [View, e.g. 3/4 angled] view. This is pure product photography: apply high-quality physical materials including reflective silver metal and soft matte white surfaces, professional studio lighting, even illumination, and sharp focus. The product must be completely isolated on a solid, clean, pure white background. You must absolutely exclude and not generate any human figures, faces, body parts, hands, mannequins, lifestyle contexts, environments, complex backgrounds, text, watermarks, or overlapping objects.

3D generation

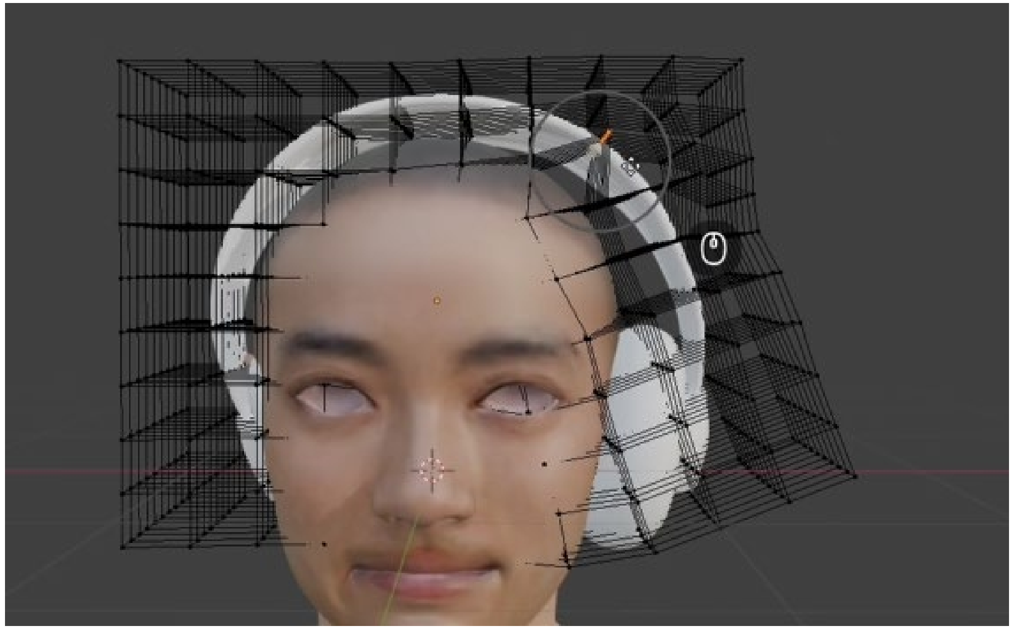
https://3d.hunyuan.tencent.com/studio/creation/role/geo

Topology

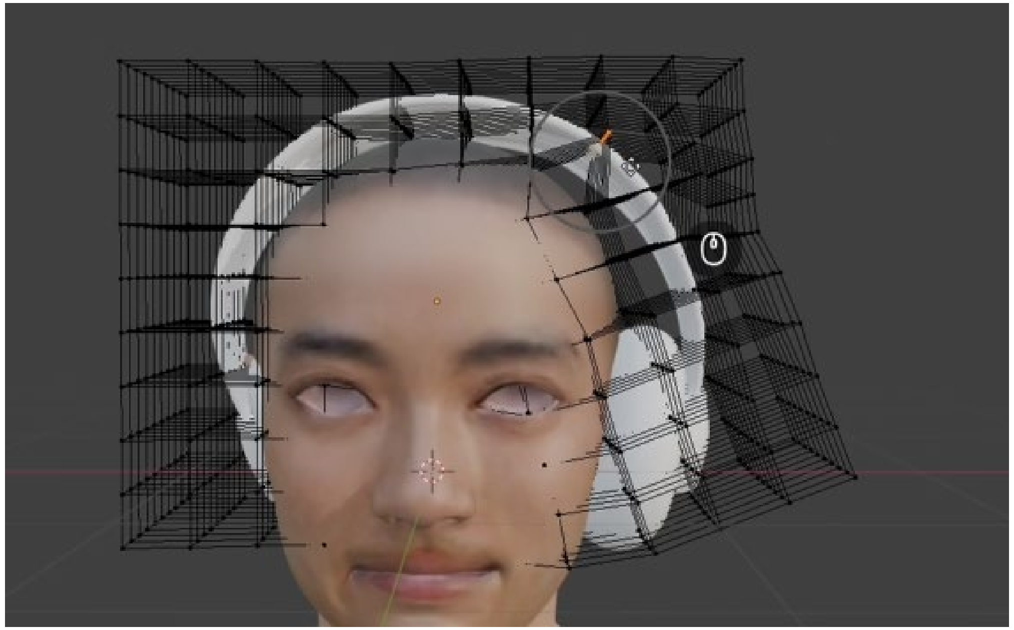
Rotate and Move

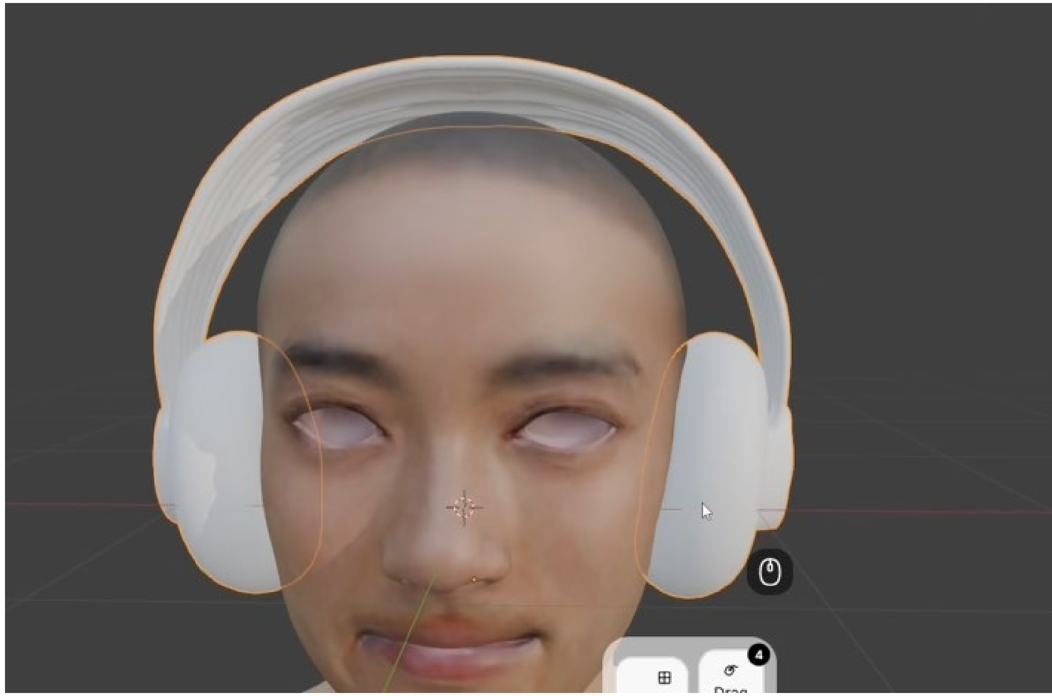


Lattice Modifier



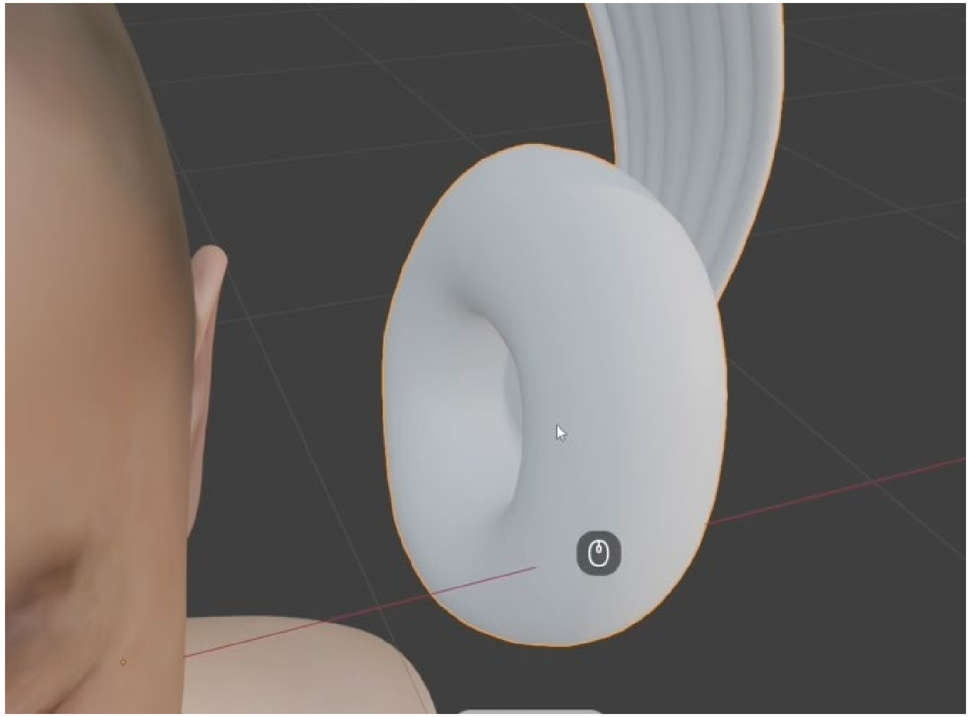
Mirror

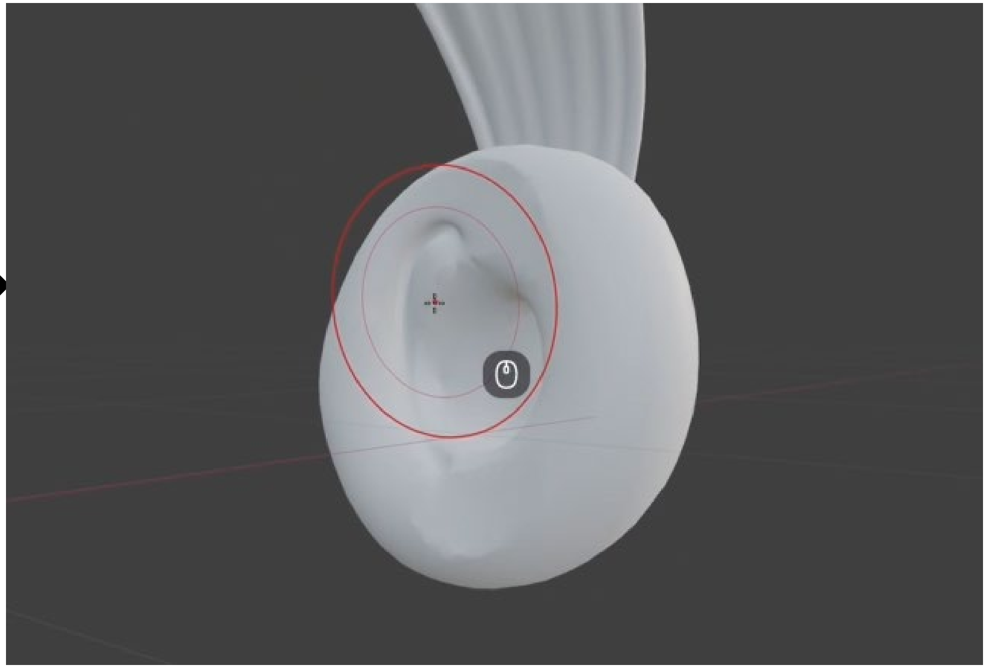




Bool

Fit to ear





Application